**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 103 · Lesson 103 · Input and Output: Selecting Devices for Tasks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does an input device do?

Q2. What does an output device do?

Q3. What is an actuator?

Q4. A 2×3 display grid: how many cells?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

## New learning

Understand — An input device captures data or a user action; an output device communicates information or changes physical conditions.

Apply — A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

Evaluate — Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

## Worked example

Recommend input and output for automated greenhouse temperature control.

1. Explain the choice — Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

# We Do · Practise with guidance

A library kiosk needs to identify books and display return confirmation. Select suitable devices.

Explain the choice

Input barcode reader or RFID reader; output display and optional printer.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

### Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]

### Question 3

Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

# Exit Ticket · Answer before checking

A touchscreen can only be an output device.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Captures user/environment data.

Q2. Communicates information or causes action.

Q3. Output device controlling physical movement/action.

Q4. Six.

### You Do pages · Independent questions

Q1. Sensor measures temperature [2], CPU compares threshold [2], actuator controls fan/heater [2].

Q2. Barcode cheap but line-of-sight [3], RFID can scan tags without direct optical sight but cost/interference [3].

Q3. Touch or keypad as input [2], audio navigation/physical buttons [2], readable display [2], spoken confirmation/printed ticket [2].

### Exit Ticket · Model answer

False. It both displays and senses touch input.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**